

Probability
Chapter 3 - Module 3

Conditional Variance

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Def. (Conditional Variance) Let X and Y be random variables. We define the conditional variance by

$$\text{Var}(X|Y = y) = E[(X - E[X | Y = y])^2 | Y = y].$$

Example 1. Suppose that we have an urn with N balls, K of which are white and the remaining $N - K$ balls are black. Suppose that we randomly select n balls (without replacement) from the urn.

Let $X_i = 1$ if we choose a white ball (success)

Let $X_i = 0$ if we choose a black ball (failure).

Let $S = \sum_{i=1}^n X_i$, which is the total number of successes. As mentioned before, this is the classical hypergeometric random variable.

Then

$$P(S = k) = \frac{\binom{K}{k} \binom{N-K}{n-k}}{\binom{N}{n}}$$

and

$$E[S] = n \frac{K}{N}$$

and

$$\text{Var}(S) = n \frac{K}{N} \frac{N-K}{N} \frac{N-n}{N-1}$$

for $k = 0, \dots, K$ and $n = 1, \dots, N$

Let's determine $E[S | X_1]$ and $\text{Var}(S | X_1)$.

Theorem. Let X and Y be random variables. Then

$$\text{Var}(X) = E[\text{Var}(X | Y)] + \text{Var}(E[X | Y]).$$

Proof. We begin by recalling that

$$\text{Var}(X) = E[X^2] - (E[X])^2.$$

We extend this to conditional variance by noting that

$$\begin{aligned}\text{Var}(X | Y = y) &= E[(X - E[X|Y = y])^2 | Y = y] \\&= E[X^2 - 2X E[X|Y = y] + (E[X|Y = y])^2 | Y = y] \\&= E[X^2 | Y = y] - 2E[X|Y = y] E[X|Y = y] + (E[X|Y = y])^2 \\&= E[X^2 | Y = y] - (E[X|Y = y])^2.\end{aligned}$$

So we conclude that

$$\text{Var}(X | Y) = E[X^2 | Y] - (E[X | Y])^2.$$

Recall that we are trying to show that

$$\text{Var}(X) = E[\text{Var}(X | Y)] + \text{Var}(E[X | Y]).$$

So let's look at each term on the right hand side.

Recall that we just showed that

$$\text{Var}(X | Y) = E[X^2 | Y] - (E[X | Y])^2$$

and so

$$\begin{aligned} E[\text{Var}(X | Y)] &= E\left[E[X^2 | Y] - (E[X | Y])^2 \right] \\ &= E\left[E[X^2 | Y] \right] - E\left[(E[X | Y])^2 \right] \\ &= E[X^2] - E[(E[X | Y])^2] \end{aligned}$$

and

$$\begin{aligned} \text{Var}(E[X | Y]) &= E[(E[X | Y])^2] - (E[E[X | Y]])^2 \\ &= E[(E[X | Y])^2] - (E[X])^2. \end{aligned}$$

If we add these terms together we see that

$$E[\text{Var}(X | Y)] + \text{Var}(E[X | Y]) = E[X^2] - (E[X])^2 = \text{Var}(X).$$

Example 2. We return to the compound random variable problem. So let $X_1, X_2, \dots$ be independent random variables with mean μ and variance σ^2 . And for convenience let X be a random variable with the same mean and variance.

Let $S = \sum_{i=1}^N X_i$ where N is a random variable with finite mean and variance.

Then as we have previously shown,

$$E[S] = E[N] E[X] = E[N] \mu.$$

We now determine the variance of S .

We will use the fact that

$$\text{Var}(S) = E[\text{Var}(S \mid N)] + \text{Var}(E[S \mid N]).$$

So note that

$$\begin{aligned}\text{Var}(S \mid N = n) &= \text{Var}\left(\sum_{i=1}^N X_i \mid 3N = n\right) \\ &= \text{Var}\left(\sum_{i=1}^n X_i\right) \\ &= \sum_{i=1}^n \text{Var}(X_i) = n\sigma^2\end{aligned}$$

and so

$$E[\text{Var}(S \mid N)] = E[N\sigma^2] = \sigma^2 E[N].$$

Furthermore

$$\text{Var}(E[S \mid N]) = \text{Var}(N\mu) = \mu^2 \text{Var}(N).$$

Therefore

$$\text{Var}(S) = \sigma^2 E[N] + \mu^2 \text{Var}(N).$$

Let's leverage what we have learned about conditional expectations.

Let E be an event and define

$$X = \begin{cases} 1 & \text{if } E \text{ occurs} \\ 0 & \text{if } E \text{ does not occur.} \end{cases}$$

$$E[X] = 1 \cdot P(E) + 0 \cdot P(E^c) = P(E).$$

Similarly

$$E[X \mid Y = y] = P(E \mid Y = y).$$

And since

$$E[X] = E[E[X \mid Y]]$$

then equivalently

$$P(E) = E[X] = E[E[X \mid Y]].$$

Example 2. Let X and Y be independent random variables.

Assume X has CDF $F_X(x)$ and pdf $f_X(x)$

and that Y has CDF $F_Y(y)$ and pdf $f_Y(y)$.

Problem. Determine $P(X \leq Y)$.

Strategies.

1. Solve the problem directly.
2. Solve the problem using conditioning.
3. Solve the problem easily using conditioning.

To the board.

Let X_1 and X_2 be binomial random variables with parameters (p, n_1) and (p, n_2) respectively. Then $X_1 + X_2$ is a binomial random variable with parameters $(p, n_1 + n_2)$.

We proved this using characteristic functions.

By induction, if $X_1, X_2, \dots, X_n$ are independent binomial random variables with X_i having parameters (p, n_i) , then $\sum_{i=1}^n X_i$ is a binomial random variable with parameters $\left(p, \sum_{i=1}^n n_i\right)$.

Instead suppose that X_i is a binomial random variable with parameters (p_i, n_i) for $i = 1, \dots, n$.

Question: What is the distribution of $Y_n = \sum_{i=1}^n X_i$?

I don't know, but consider this.

Example 3. For $i = 1, \dots, n$ let X_i be binomial random variables with parameters (p_i, n_i) . And suppose that $X_1, X_2, \dots, X_n$ are independent.

Define

$$P_k(j) = P(X_1 + X_2 + \dots + X_k = j)$$

for

$$0 \leq j \leq \sum_{i=1}^k n_i \equiv N_k.$$

And for convenience define $q_i = 1 - p_i$.

We know that $P_1(j) = P(X_1 = j) = \binom{n_1}{j} p_1^j q_1^{n_1-j}$ for $j = 0, \dots, n_1$.

In addition

$$\begin{aligned} P_k(j) &= P(X_1 + \dots + X_k = j) \\ &= \sum_{i=0}^{n_k} P(X_1 + \dots + X_k = j \mid X_k = i) P(X_k = i) \\ &= \sum_{i=0}^{n_k} P(X_1 + \dots + X_{k-1} = j - i) P(X_k = i) \\ &= \sum_{i=0}^{n_k} P_{k-1}(j - i) P(X_k = i), \end{aligned}$$

for $k = 2, \dots, n$ and where $P_1(j)$ is given above and where $P_l(j) = 0$ when $j < 0$.

This recursively defines the numbers $P_k(j)$.